Databases

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Governments systematically collect and record data about the population through a process called Census. Census data contains valuable information which are helpful in planning and formulating policies.

Data are indeed crucial for decision making.

In general, data is a collection of characters, numbers, and other symbols that represents values of some situation or variable. Data is plural and singular of the word data is "datum". Using computers, data are stored in electronic forms because data processing becomes faster and easier as compared to manual data processing done by people. The information and communication technology (ICT) revolution led by computer, mobile and Internet has resulted. In generation of large volume of data at a very fast pace. The following list contains some examples of data that we often come across:

- Name, age, gender, contact details, etc. of a person
- Transaction data generated through banking, Ticketing, shopping, etc. whether online or offline.
- Documents and web pages.
- Online posts, comments and messages.
- Signal generated by sensors
- Satellite data, including meteorological data, communication data, earth observation data, etc.

Importance of Data

Type of Data

As data comes from different sources, they can be in different formats. For example, an image is a collection of pixels. A video is made up of frames. A fee slip is made up of few numeric and non-numeric entries. And messages or chats are made up of text, icons or emoticons or emojis and images or videos. Two broad categories in which data can be classified on the basis of their format are:

A. **Structured Data:** Data which is organized and can be recorded in a well-defined format is called structured data. Structure data is usually stored in computer in a tabular (in rope and columns) format, where each column represents different data for a particular parameter called attribute or characteristics or variable and each row represents data of an observation of different attributes.

For example: Structured data about kitchen items in a shop

ModelNo	Product_Name	Unit_Price	Discount%	Items _in_inventory
ABC1	Water bottle	126	8	13
ABC2	Melamine Plate	320	5	45
ABC3	Dinner Set	4200	10	10
GH67	Jug	80	0	15
GH78	Table Spoon	120	5	100
Gh81	Bucket	190	12	10
NK2	Kitchen Towel	25	0	50

Attributes maintained for different Activities

Entity/Activities	**Data Fields/Parameters/Attributes**
Books at a shop	**Book_title, Author, Price, Year_of_publication**

Depositing fee in a school	**Student_name, class, roll_number, FeeAmount, Deposit_date**
Amount withdrawal from ATM	**AccHolderName, AccountNumber, TypeOfAccount, DateOfWithdrawl, AmountWithdrawn, ATMid, TimeOfWithdrawl**

B. Unstructured Data: A newspaper contains various types of news items which are also called data. But there is no fixed pattern that newspaper follows in placing news articles. One day there might be three images of different sizes on a page, along with a 5 News items and one or more advertisements. On another day might be one big image with three textual news items. So there is no particular format nor any fixed structure for printing news. Another example is the content of an email. There is no fixed. Structure about how many lines or paragraphs 1 has to write in an email or how many files are to be attached with an email. In summary, data which are not in the traditional row and column structure is called unstructured data.

Example of unstructured data include web pages consisting of text as well as multimedia contents (images, graphics, audio or video). Other examples include text documents, business reports, books, audio or video files, social media messages. Although there are ways to process unstructured data, we are going to focus on handling structured data only.

Unstructured data are sometimes described with the help of some other data called **metadata**. Metadata is basically data about data. For example, we describe different parts of an email as subject, recipient, main body, attachment, etc. These are the metadata for the email data. Likewise, we can have some metadata for an image file, such as image size (in KB or MB) Image type, for (example JPEG, PNG), image resolution, etc.

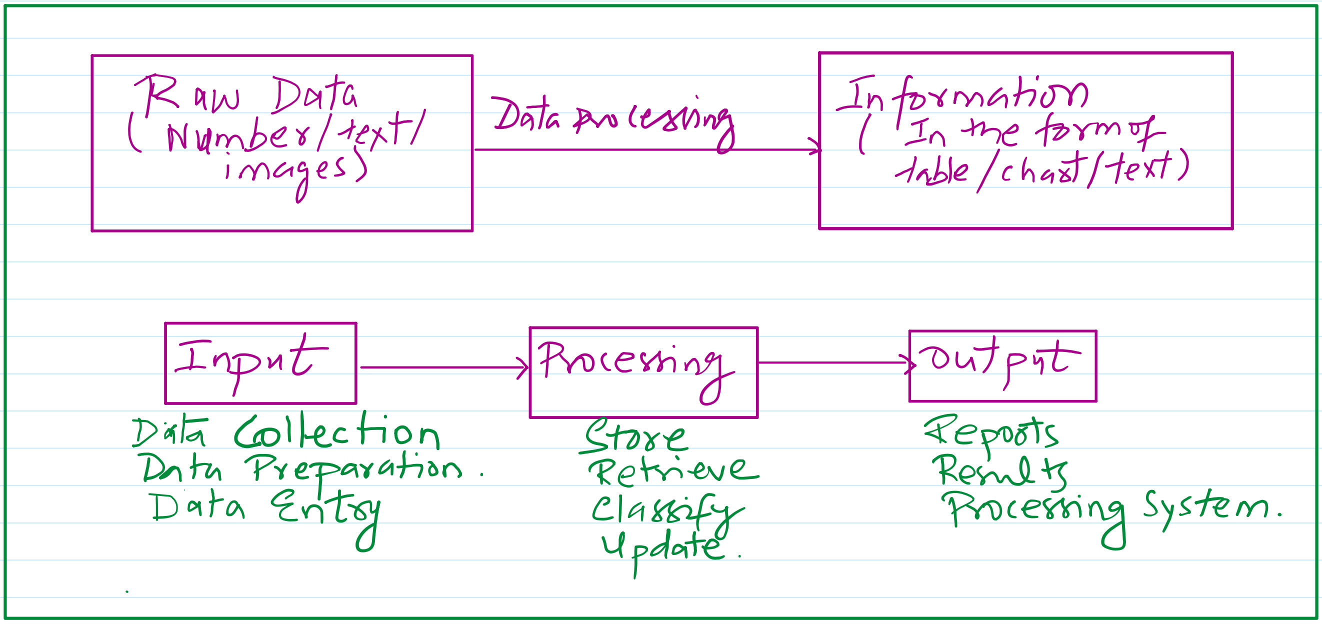
Data Collection

For processing data we need to collect or gather data first. We can then store the data in a file or database for later use. Data collection here means identifying already available data or collecting from the appropriate sources. Suppose there are three different scenarios where sales data in a grocery stores are available:

- Sales data are available with the shopkeeper in a diary or register. In this case we should enter the data in a digital format, for example in a spreadsheet.
- Data already available in digital format, say in a CSV(comma separated values) file.
- The shopkeeper has so far not recorded any data in either form, but wants to get a software developed for maintaining sales data and accounts. The software may be developed using programming language such as Java or Python. Which can be used to store and retrieve data from a CSV file or a database management system like MySQL.

Data storage: Once we gather data and process them to get results, we may not then simply discard the data. Rather, we would like to store them for future use as well. Data storage is a process of storing data on storage devices so that data can be retrieved later. Nowadays large volume of data are being generated at very high rate. As a result, data storage has become a challenging task. However, the decrease in the cost of digital storage devices has helped in simplifying this task. There are numerous digital storage devices available in the market. Hard disk drive (HDD closed parenthesis. Solid state drive (SSD) Pen drive. Flash drives. Memory cards, etc.
We store data like images, documents, audios or videos etcetera as files in our computer. Likewise, school or hospital data are stored in data files. We use computers to add, modify or delete data in these files or process these data files to get results. However, file processing has certain limitations which can be overcome through database management system. (DBMS)

Data Processing: We are interested in understanding data as they hold valuable facts and information that can be useful in our decision making process. However, by looking at the vast or large amount of data, one cannot arrive at a conclusion. Rather, data needs to be processed to get results and after analyzing those results we can make conclusions or take decisions.
We find automated data processing in situations like one bill payment. Registration of complaints, booking tickets, etc.



Steps in Data Processing.

Statistical Techniques For Data Processing.

Given a set of data values, we need to process them to get information. There are various techniques which help us to have preliminary understanding about the data. Summarization methods are applied on tabular data for its easy comprehension. Commonly used statistical techniques for data summarization are given below:

Measure of central tendency
A major of central tendency is a single value that gives us some idea about the data. Three most common measures of central tendency are **mean, median and mode**. Instead. Of looking at each individual data values. We can calculate the mean, median and mode of the data to get an idea about average middle value and frequency of occurrence of a particular value respectively. Selection of a measure of central tendency depends on certain characteristics of data.

$$M = \left[\sum_{i}^{n} x_i\right] / n .$$

Median : 1. Sort the data [ascending/descending].
if n is odd then select $\frac{n}{2}$ th element.
else n is even then select. (n & n+1)th element and divide by 2.

else n is even then select (n & n+1)-th element and divide by 2.

Mode: we find the element, which has appeared for most number of times.

Measure of Variability: The measure of variability refers to the spread or variation of the values around the mean. They are also called measures of dispersion that indicate the degree of diversity in a data set. They also indicate difference within the group. Two different data sets can have the same mean, median, or mode, but completely different levels of dispersion, or vice versa. Common measures of dispersion are variability are range and standard deviation.

Range: It is a difference between maximum and minimum values of the data (the largest value minus the smallest value) Range can be calculated only for numerical data. It is a measure of dispersion and tells about coverage or spread of data values. For example, difference in salaries of employees, marks of student, price of toys etc. As range is calculated based on two extreme values. Any outlier in the data badly influences the result.
Let M be the largest maximum value and S is the smallest or minimum value in the data. Then range is the difference between two extreme values, that is (M-S) or (maximum -minimum).

Standard deviation:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n}}$$

Height (x) in cm	$x - \bar{x}$	$(x - \bar{x})^2$
90	-11.33	128.37
102	0.67	0.45
110	8.67	75.17
115	13.67	186.87
85	-16.33	266.67
90	-11.33	128.37
100	-1.33	1.77
110	8.67	75.17
110	8.67	75.17
$n = 9$, $\bar{x} = 101.33$	$\sum(x - \bar{x}) = 0.03$	$\sum(x - \bar{x})^2 = 938.01$

$$\sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}} = \sqrt{\frac{938.01}{9}} \approx \sqrt{104.11} \approx \underline{10.2 \text{ cm.}}$$

128.37 + 0.45 + 75.17 + 186.87 +266.67+128.37+ 1.77+75.17+75.17 = 938.01

8.67 *8.67 = 75.1689
13.67 * 13.67 = 186.8689
16.33 ^ 2 = 266.6689
11.33 ^2 = 128.3689
1.33 ^2 = 1.7689

Problem statement	**Choose suitable statistical method.**

The management of a company wants to know about disparity in salaries of all employees.	
Teacher wants to know about the average performance of the whole class in a test.	
Find the dominant value from a set of values.	
Compare income of residence of two cities.	
Find the popular color of a car after surveying the car owners of a small city.	

Next class Database concepts